UML Diagrams

bsi-test-dashboard

Black Sphere Industries

Generated 2026-03-23

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[text] =

[OK] SEQUENCE DIAGRAM: Submit Test Run (POST /api/runs)

[OK] SEQUENCE DIAGRAM: GitHub Webhook Processing

[OK] SEQUENCE DIAGRAM: Dashboard Initialization and Data Loading

[OK] SEQUENCE DIAGRAM: Remote Test Runner Execution

[text] COMPONENT DIAGRAM

[text] STATE DIAGRAM

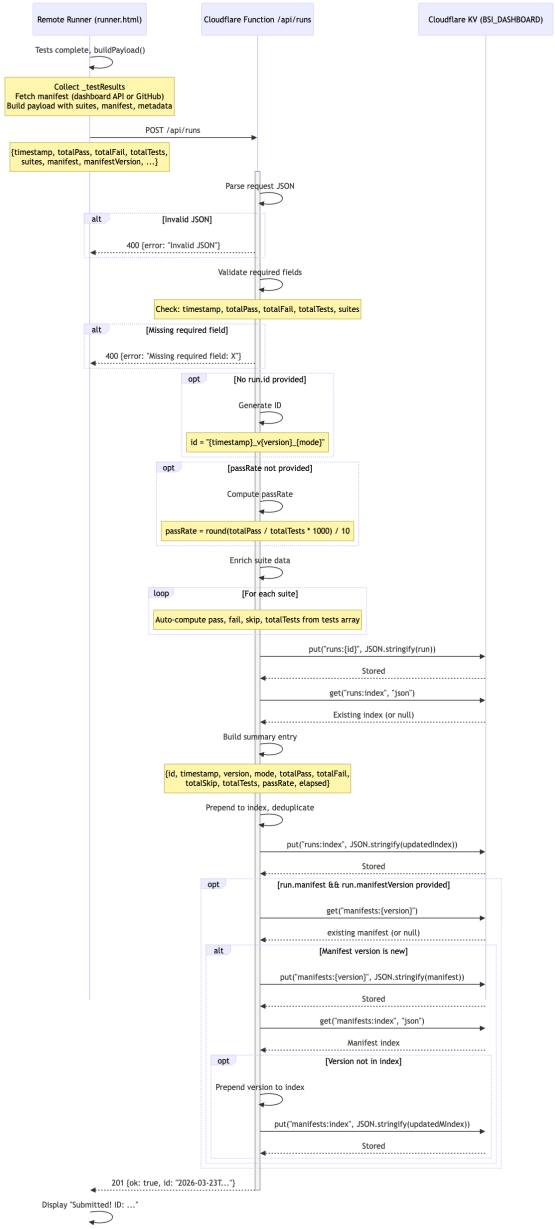
=

```
classDiagram
    class DashboardApp {
        -Object runsIndex
        -Map runCache
        -Array events
        -Object manifests
        -String runsSortKey
        -Boolean runsSortAsc
        -String runsFilterMode
        -Set selectedRuns
        -Map charts
        +initDashboard()
        +loadAll() Promise
        +showSection(id, el)
        +toggleTheme()
    }

    class AuthModule {
        -Object FIREBASE_CONFIG
        -Object fbApp
        -Object auth
        -Object googleProvider
        +signInWithGoogle()
        +signOut()
        -onAuthStateChanged(user)
    }

    class RunsIndex {
        +String lastUpdated
        +Array~RunSummary~ runs
    }

    class RunSummary {
        +String id
        +String timestamp
        +String version
        +String mode
        +Number totalPass
        +Number totalFail
        +Number totalSkip
        +Number totalTests
        +Number passRate
        +Number elapsed
    }

    class TestRun {
        +String id
        +String timestamp
        +String mode
        +String runner
        +String hostname
        +Boolean completed
        +Number totalPass
        +Number totalFail
        +Number totalSkip
        +Number totalTests
        +Number elapsed
```

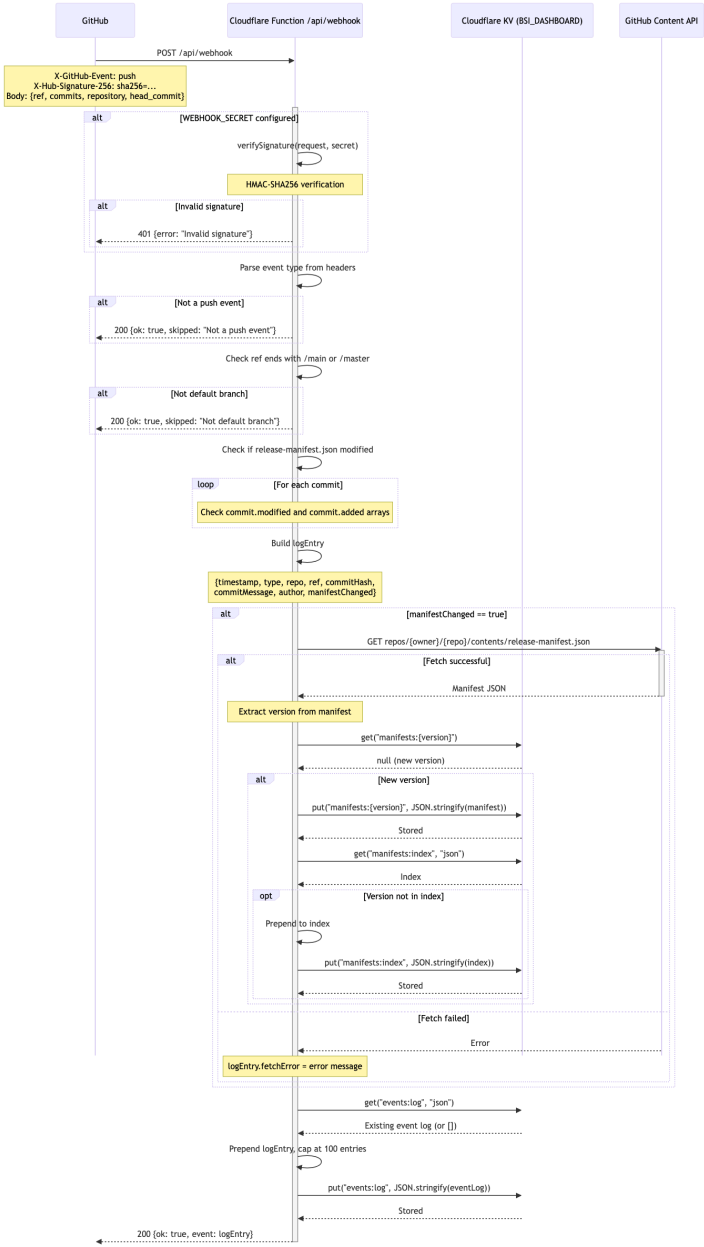
```
        +Number passRate
        +Array~SuiteResult~ suites
        +Object manifest
        +String manifestVersion
        +String manifestDate
    }

    class SuiteResult {
        +String name
        +Number totalTests
        +Number pass
        +Number fail
        +Number skip
        +Array~TestDetail~ tests
    }

    class TestDetail {
        +String name
        +String status
        +String error
        +Number elapsed
    }
```

... (260 lines total - see .mmd source)

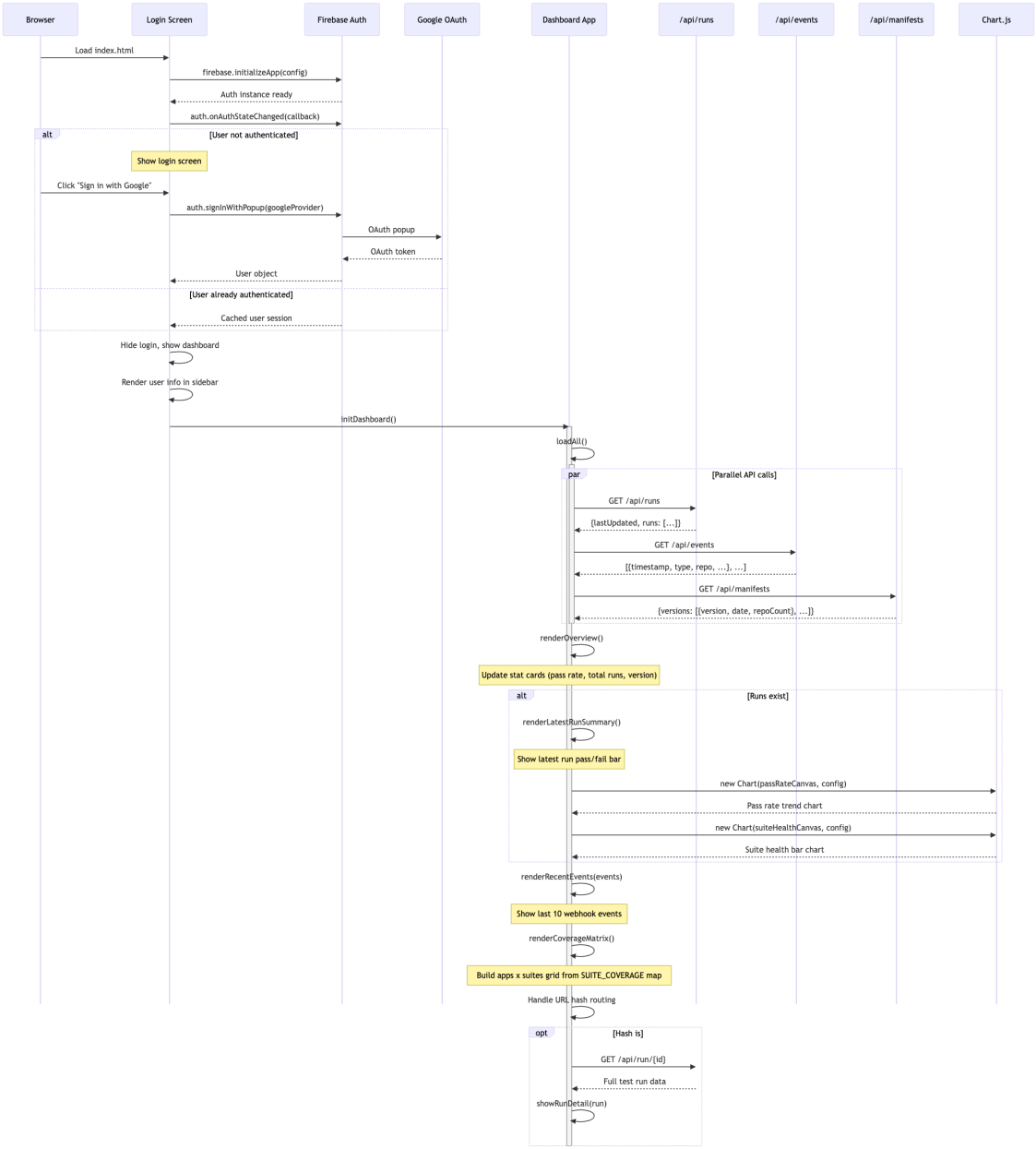
SEQUENCE DIAGRAM: Submit Test Run (POST /api/runs)



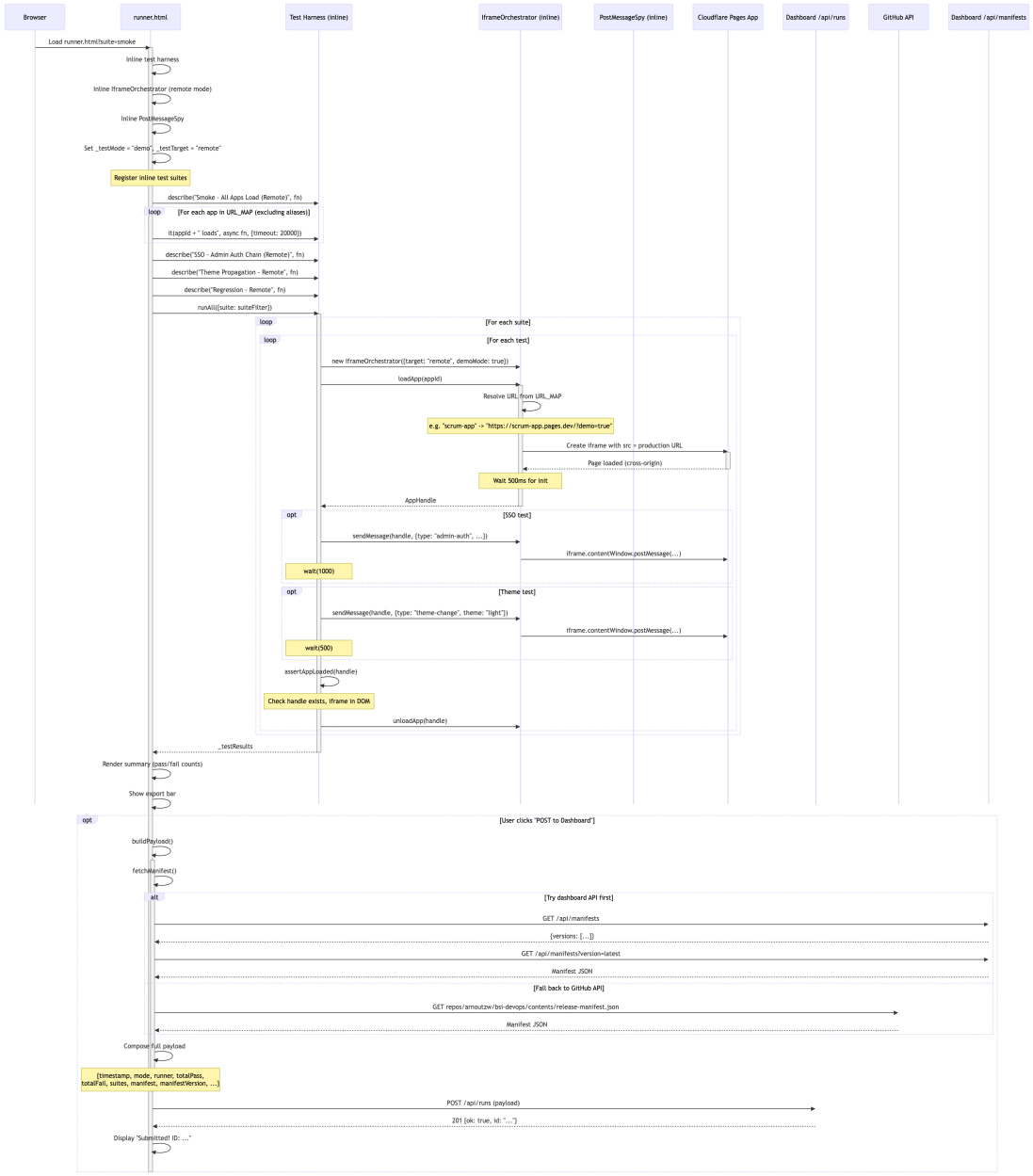
SEQUENCE DIAGRAM: GitHub Webhook Processing



SEQUENCE DIAGRAM: Dashboard Initialization and Data Loading



SEQUENCE DIAGRAM: Remote Test Runner Execution



COMPONENT DIAGRAM

```
graph TB
    subgraph "BSI Test Dashboard"
        subgraph "Frontend (index.html)"
            LOGIN[Login Screen<br/>Firebase Google Auth]
            SIDEBAR[Sidebar Navigation<br/>Overview, Runs, Detail, Manifest,<br/>Coverage, Trends, Settings]

            subgraph "Dashboard Sections"
                OVERVIEW[Overview<br/>Stats + Charts + Events]
                RUNS_LIST[Runs List<br/>Sortable + Filterable Table]
                RUN_DETAIL[Run Detail<br/>Suite Breakdown + Tests]
                MANIFEST_VIEW[Manifest Explorer<br/>Version History + Repos]
                COVERAGE[Coverage Map<br/>Apps x Suites Matrix]
                TRENDS[Trends<br/>Rate/Duration Charts + Heatmap<br/>+ Flaky Detection]
                SETTINGS[Settings<br/>Import + Webhook Setup]
            end

            subgraph "JS Libraries (CDN)"
                TAILWIND[Tailwind CSS]
                CHARTJS[Chart.js]
                LUCIDE[Lucide Icons]
                FIREBASE_AUTH[Firebase Auth SDK]
            end
        end

        subgraph "Remote Runner (runner.html)"
            RUNNER_HARNESS[Inline Test Harness<br/>describe/it/assert]
            RUNNER_ORCH[Inline IframeOrchestrator<br/>Remote Mode]
            RUNNER_SPY[Inline PostMessageSpy]
            RUNNER_SUITES[Test Suites<br/>Smoke + SSO + Theme + Regression]
            EXPORT[Export/Submit Controls]
        end

        subgraph "Cloudflare Functions (API)"
            FN_RUNS[/api/runs<br/>GET list, POST submit]
            FN_RUN_ID[/api/run/:id<br/>GET detail, DELETE]
            FN_EVENTS[/api/events<br/>GET webhook log]
            FN_MANIFESTS[/api/manifests<br/>GET index, GET by version]
            FN_WEBHOOK[/api/webhook<br/>POST GitHub webhook]
        end

        subgraph "Shell Scripts"
            SUBMIT_SH[submit-run.sh<br/>CLI Run Submission]
        end
    end

    subgraph "Infrastructure"
        CF_KV[(Cloudflare KV<br/>BSI_DASHBOARD namespace)]
        CF_PAGES[Cloudflare Pages<br/>Static Hosting + Functions]
    end

    subgraph "External"
        FIREBASE[Firebase Auth<br/>arnout-s-homelab]
        GITHUB_WH[GitHub Webhooks<br/>bsi-devops repo]
        GITHUB_API[GitHub Content API<br/>release-manifest.json]
        BSI_APPS[BSI Production Apps<br/>30 Cloudflare Pages sites]
    end
```

```
    LOGIN --> FIREBASE_AUTH
    FIREBASE_AUTH --> FIREBASE

    OVERVIEW --> FN_RUNS
    OVERVIEW --> FN_EVENTS
    OVERVIEW --> FN_MANIFESTS
    RUNS_LIST --> FN_RUNS
    RUN_DETAIL --> FN_RUN_ID
    MANIFEST_VIEW --> FN_MANIFESTS
    SETTINGS -.->|Import JSON| FN_RUNS

    OVERVIEW --> CHARTJS
    TRENDS --> CHARTJS

    RUNNER_ORCH --> BSI_APPS
    EXPORT --> FN_RUNS
    EXPORT -.->|fetch manifest| FN_MANIFESTS
    EXPORT -.->|fallback| GITHUB_API

    FN_RUNS --> CF_KV
    FN_RUN_ID --> CF_KV
    FN_EVENTS --> CF_KV
    FN_MANIFESTS --> CF_KV
```

... (92 lines total - see .mmd source)

STATE DIAGRAM

```
stateDiagram-v2
    state "Dashboard App States" as DashboardStates {
        [*] --> Unauthenticated : Page loaded

        state Unauthenticated {
            [*] --> ShowLoginScreen
            ShowLoginScreen --> AuthPending : Click Sign In
            AuthPending --> ShowLoginScreen : Auth cancelled / error
        }

        Unauthenticated --> Authenticated : Firebase auth success

        state Authenticated {
            [*] --> LoadingData
            LoadingData --> DataReady : API calls complete

            state DataReady {
                [*] --> OverviewView
                OverviewView --> RunsView : Click Runs nav
                RunsView --> DetailView : Click run row
                DetailView --> RunsView : Click Back
                OverviewView --> ManifestView : Click Manifest nav
                OverviewView --> CoverageView : Click Coverage nav
                OverviewView --> TrendsView : Click Trends nav
                OverviewView --> SettingsView : Click Settings nav
                RunsView --> OverviewView : Click Overview nav
                ManifestView --> OverviewView : Click Overview nav
                CoverageView --> OverviewView : Click Overview nav
                TrendsView --> OverviewView : Click Overview nav
                SettingsView --> OverviewView : Click Overview nav
            end
        end

        Authenticated --> Unauthenticated : Sign out
    }

    state "Test Run Lifecycle" as TestRunLifecycle {
        [*] --> RunCreated : Runner generates results

        state RunCreated {
            [*] --> LocalResults
            LocalResults --> PayloadBuilt : buildPayload()

            state PayloadBuilt {
                [*] --> FetchingManifest
                FetchingManifest --> ManifestFromDashboard : Dashboard API has manifest
                FetchingManifest --> ManifestFromGitHub : Fallback to GitHub API
                FetchingManifest --> NoManifest : Both sources fail
                ManifestFromDashboard --> PayloadReady
                ManifestFromGitHub --> PayloadReady
                NoManifest --> PayloadReady : manifest = null
            }
        end

        PayloadReady --> Submitting : POST to /api/runs
        Submitting --> StoredInKV : 201 response
        Submitting --> SubmitFailed : Error response
```

```
        StoredInKV --> IndexUpdated : Index prepended
        IndexUpdated --> ManifestSnapshotted : New manifest version stored

        state ManifestSnapshotted {
            [*] --> CheckExisting
            CheckExisting --> AlreadyExists : Version known
            CheckExisting --> StoreNewManifest : New version
            StoreNewManifest --> UpdateManifestIndex
        }

        ManifestSnapshotted --> RunComplete : All stored

        RunComplete --> [*]
        SubmitFailed --> [*]
    }

    state "Webhook Event Processing" as WebhookStates {
        [*] --> EventReceived : GitHub sends POST

        EventReceived --> VerifyingSignature

        state VerifyingSignature {
```

... (138 lines total - see .mmd source)